

Transaction Fraud Detection Method Based on Event Temporal Graph Attention Network

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Abstract

In recent years, financial crimes such as telecom network fraud have shown highly dynamic characteristics. Criminals implement money laundering through complex fund transfer paths, posing severe challenges to anti-money laundering supervision. Traditional account-based graph construction methods are limited in detecting such abnormal transactions, as they fail to capture the temporal order of transactions and the topological anomalies of fund transfers. To address this, a transaction fraud detection model based on event temporal graph attention network (ETGAT) is proposed. This model constructs graphs with transaction events as nodes, enhances node features through time encoding, and captures the global information of the graph using a Graph Attention Network (GAT) layer. Experimental results show that ETGAT exhibits good performance and interpretability, providing a new solution for transaction fraud detection.

CCS Concepts

• Applied computing; • Electronic commerce; • Secure online transactions;

Keywords

Transaction fraud detection, Event temporal, Graph attention network, Time encoding

ACM Reference Format:

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1 Introduction

In recent years, new types of telecom network fraud have shown highly dynamic characteristics, specifically manifested in property fraud by impersonating public institutions, phishing attacks by mimicking e-commerce customer service, and spreading malicious

programs using pseudo-base station technology. With the evolution of information technology, these criminal activities have developed into systematic risk sources, posing severe challenges to national financial security and social credit systems. In terms of abnormal fund transfer features, criminals implement fund transfers through multi-layer nested user transaction paths to obscure illegal fund sources and achieve the legalization of illicit funds through money laundering. This type of financial crime has become a key governance focus in current anti-money laundering supervision.

Money laundering activities are based on transactions, and there are numerous studies on transaction fraud both domestically and internationally. Liu Yan [1] et al. proposed using a deep learning model combining autoencoders and restricted Boltzmann machines for detecting credit card transaction fraud, which can reconstruct normal transactions and identify abnormalities from normal patterns. Wang [2] applied a deep forest model for online transaction fraud detection, generating individual credibility and group anomaly scores based on transaction times to classify legitimate and fraudulent transactions. Zhang [3] proposed a federated meta-learning framework for credit card fraud detection, enabling banks to fully utilize locally stored training data to train fraud detection models. With the development of deep learning, Liu [4] et al. considered node attributes and proposed an attention-based graph representation learning model, achieving good results. Chen [4] et al. proposed a financial fraud detection scheme based on deep convolutional neural networks (DCNN), comparing their proposed model with existing machine learning models and deep learning models like autoencoders to evaluate its performance. Zhang X [5] developed a fraud detection system using a deep learning architecture and advanced feature engineering based on homogeneity behavior analysis to handle credit card fraud detection problems. Xie [6] et al. proposed the TH-LSTM model, improving long short-term memory and time-aware gates to capture behavioral changes caused by users' consecutive transactions. Xu [7] et al. proposed TGAT, an algorithm that combines neighbor node features and corresponding time information to represent nodes iteratively on temporal graphs. TGAT developed a set of time encoding functions based on Bochner's theorem to characterize time difference information. Rao [8] et al. proposed dynamic heterogeneous graph neural networks (DyHGN) and its variants to capture the dynamics of graphs, combining attention mechanisms and time encoding to significantly improve fraud detection performance. Wang [9] et al. proposed a neural network model specifically designed for real-time temporal graph embedding, decoupling model inference and graph computation to greatly increase real-time inference speed.

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Nguyen [10] proposed CTDNE, a general framework for incorporating time dependency into existing random walk-based network embedding methods, capable of learning time embeddings from continuous-time dynamic networks. Zou [11] et al. modeled dynamic interactions and time dependencies between nodes using Hawkes processes, effectively capturing the dynamic changes in time networks. Zhang [12] et al. proposed TigeCMN, a new framework for learning node representations from a series of time interactions, using two coupled memory networks to store and update node embeddings, forming deep matrix representations, and enhancing expressiveness. Abbas [13] proposed a new time network embedding algorithm, generating low-dimensional embeddings by considering historical interactions and time order between nodes. These embeddings better capture the dynamic characteristics of nodes, aiding future link prediction. Yu [14] et al. proposed a novel dynamic graph learning architecture and unified library, providing new tools and methods for dynamic graph learning. Wang [15] et al. proposed CAW (Causal Anonymous Walks), extracting time network patterns via time random walks and avoiding the cumbersome process of manually selecting and counting patterns. CAW adopts a novel anonymization strategy, replacing node identities with their occurrence counts in sampled walks, maintaining inductivity while preserving pattern correlations.

The core risk characteristics of money laundering lie in the topological anomalies of fund transfer networks. Existing research mainly adopts two network modeling paradigms: homogeneous transaction network models based on user nodes and heterogeneous information network models integrating multidimensional features. The former establishes a user-based sequential transaction association network, using high-frequency transaction features and asymmetric fund flows as edge weight parameters. Studies show that the predictive efficiency of this paradigm is limited by the completeness of edge weight definitions. When faced with bidirectional or composite transaction scenarios, the accuracy of risk identification significantly decreases. The latter enhances feature representation capabilities by introducing diverse entity nodes such as device fingerprints but increases network topology complexity, significantly impacting algorithm performance, especially when processing large-scale dynamic transaction data, leading to computational efficiency bottlenecks. Despite the good performance of graph models in financial fraud tasks, the following challenges remain:

1. Long graph construction time, typically based on multi-day accumulated transaction data, forming static graphs. For fast-in-fast-out money laundering scenarios, it is difficult to detect anomalies promptly.
2. Current graph construction methods are user-centric, making it challenging to determine whether users are fraudulent based on hidden abnormal transactions among numerous normal transactions. Explaining users' fraudulent behaviors from a model perspective is also difficult.

To address these issues, this paper proposes a transaction fraud detection model based on an event temporal graph attention network (ETGAT), constructing graphs based on transaction events in chronological order and enhancing node features through time encoding, then introducing a Graph Attention Network (GAT) layer to

capture global graph information. Experiments were conducted using the proposed method, compared with existing methods, proving that our method has better performance.

2 Transaction fraud detection model based on an event temporal graph attention network

Next, we will introduce the algorithm process in detail.

2.1 Graph Construction

Graph structure is a data structure describing relationships between entities, generally represented as $G=(V,E)$, where V represents the set of vertices, and E represents the set of edges. Traditionally, user-centric graph construction is used in transaction fraud detection. However, this paper employs event-centric graph construction. Two graph construction methods are illustrated in Figure 1. Here, e_i represents transaction event i , u_{i1} , u_{i2} represent the active and passive users in transaction event i , and t_i represents the time of transaction event i .

Based on event-centric construction logic:

- **Nodes:** Events as nodes, with active and passive user features, event features as node attributes.
- **Edges:** Funding transfer relationships between events as edges. An edge exists from event i to event j if and only if:
 - The passive party of event i is the active party of event j .
 - Event i occurs before event j .
 - Event j occurs within 24 hours after event i .

In this scheme, a transaction event is a node, e.g., a transaction e_i between user u_{i1} and user u_{i2} forms one node, with node features including u_{i1} and user u_{i2} 's attributes, the amount, time, event name, etc. Event-based funding transfer relationships form directed, featureless edges. For example, if user u_{i1} receives money as a passive party in event e_i and subsequently sends money as an active party in event e_j , there is a directed edge from e_i to e_j .

From Figure 1, it is clear that user-based graph construction lacks the sequential relationship information of multiple transactions, which is crucial in transaction fraud, especially money laundering scenarios. Event-based graph construction captures the temporal sequence of events, facilitating discovery and explanation of model predictions.

2.2 ETGNN Model Design

The event temporal graph neural network aims to directly model temporal events, incorporating time and event sequence information through time encoding to learn these features.

2.2.1 Time Encoding. We aim to incorporate transaction event timestamps into node features to enhance feature expression. Absolute time has little significance; relative time between two nodes better expresses temporal interaction. Considering the inner product of two time nodes and their time encodings, we seek a kernel function $K(t_1, t_2)$ satisfying:

$$K(t_1, t_2) = \psi(t_1, t_2) \quad (1)$$

$$K(t_1 + c, t_2 + c) = \psi(t_1 - t_2) = K(t_1, t_2) \quad (2)$$

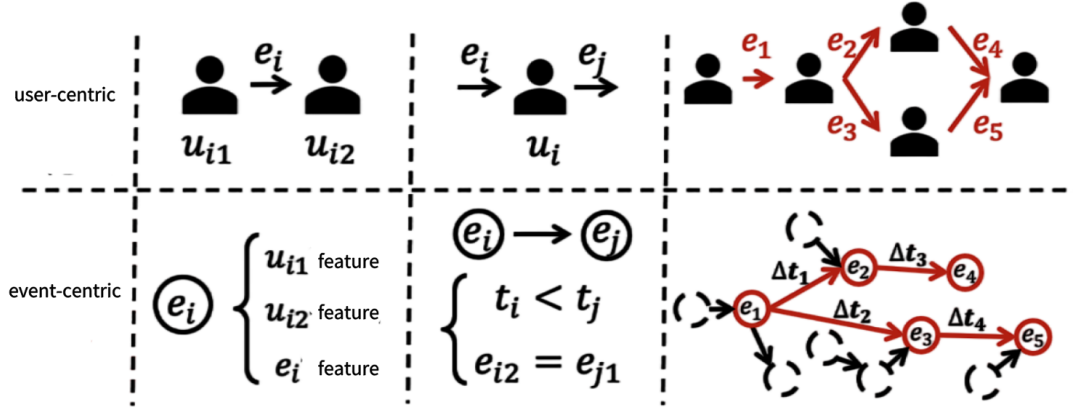


Figure 1: Two Topological Structures of Graph Construction

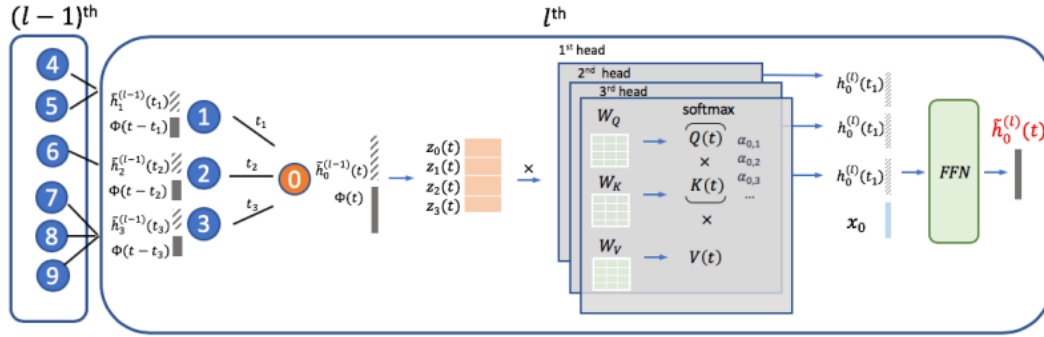


Figure 2: Event Temporal GAT Layer Algorithm

where c is a constant. According to Bochner's theorem, the kernel function can be expanded as:

$$\begin{aligned} K(t_1, t_2) &= \psi(t_1, t_2) \\ &= \int_{\mathbb{R}} e^{i\omega(t_1 - t_2)} p(\omega) d\omega = E_{\omega} [\xi_{\omega}(t_1) \xi_{\omega}(t_2)] \end{aligned} \quad (3)$$

where $\xi_{\omega}(t) = e^{i\omega t}$, since both the kernel function K and the probability $p(\omega)$ here are real numbers, then only the real part is taken to obtain the translation, we get:

$$\begin{aligned} K(t_1 - t_2) &= E_{\omega} [\cos(\omega(t_1 - t_2))] \\ &= E_{\omega} [\cos(\omega t_1) \cos(\omega t_2) + \sin(\omega t_1) \sin(\omega t_2)] \end{aligned} \quad (4)$$

Using Monte Carlo integration to approximate this expectation, we get:

$$K(t_1 - t_2) \approx \frac{1}{d} \sum_{i=1}^d \cos(\omega_i t_1) \cos(\omega_i t_2) + \sin(\omega_i t_1) \sin(\omega_i t_2) \quad (5)$$

Thus, the time feature mapping $\phi(t)$ is defined as:

$$\Phi_d(t) = [\cos(\omega_1 t), \sin(\omega_1 t), \dots, \cos(\omega_d t), \sin(\omega_d t)] \quad (6)$$

This mapping transforms time t into a 2d-dimensional vector space, each dimension corresponding to a cosine or sine function

of a random variable. This allows time information to be encoded as points in vector space, usable alongside other feature vectors in self-attention mechanisms. $\omega_1, \dots, \omega_d$ can be optimized as model parameters, initialized randomly with $d=100$.

Time feature mappings provide a method to integrate time information into node representations, enabling the model to better capture dynamic graph time features and interaction patterns.

2.2.2 GAT Layer. After adding time-encoded information to nodes, the Graph Attention Network (GAT) algorithm is applied. The network structure is shown in Figure 2.

Specific steps are as follows:

1. Input

Target Node: Assume the target node is v_0 at time t and v_0 with 120-d features.

Neighboring nodes: The neighboring node set of the target node is $N(v_0; t) = \{v_0, v_1, \dots, v_n\}$, with each neighboring node interacting with the target node at time t_i , and $t_i < t$.

Hidden representations: For the l -th layer, inputs include neighboring node hidden representations $\{h_1^{l-1}(t), \dots, h_n^{l-1}(t_n)\}$ and target node hidden representation $h_0^{l-1}(t)$.

2. Feature Matrix Construction

Concatenate the hidden representations of the target node and neighboring nodes with time encodings to form the feature matrix $Z(t)$. Specifically, for each neighboring node v_0 , concatenate its hidden representation $\tilde{h}_0^{(l-1)}(t)$ with time encoding $\Phi_{dT}(t - t_i)$:

$$Z(t) = [\tilde{h}_0^{(l-1)}(t) \parallel \Phi_{dT}(0), h_0^{l-1}(t_1) \parallel \Phi_{dT}(t - t_1), \dots, h_0^{l-1}(t_n) \parallel \Phi_{dT}(t - t_n)]^T \quad (7)$$

3. Linear Projection

Apply three different linear projections to the feature matrix to obtain "query" (Q), "key" (K), and "value" (V):

$$q(t) = Z(t) W_Q, k(t) = Z(t) W_K, v(t) = Z(t) W_V \quad (8)$$

where $W_Q, W_V, W_K \in \mathbb{R}^{(d+dT) \times dh}$ are weight matrices capturing interactions between node features and time encodings.

4. Attention Weights Calculation

Use dot-product attention mechanism to calculate attention weights α_i , indicating the importance of each neighboring node to the target node:

$$\alpha_i = \exp(q^T K_i) / \left(\sum_q \exp(q^T K_q) \right) \quad (9)$$

where K_i is the "key" vector of the i -th neighboring node. Attention weights α_i reflect the influence of neighboring node v_i on the target node v_0 at time t .

5. Aggregate Neighbor Information

Weighted sum of neighboring node "value" vectors v_0 according to attention weights α_i to get aggregated neighbor representation:

$$h(t) = \text{Attn}(q(t), K(t), V(t)) = \sum_{i=1}^N \alpha_i V_i \quad (10)$$

This step aggregates neighboring node information into the target node representation, considering time information and feature interactions.

6. Combine Target Node Features

Concatenate aggregated neighbor representation with the target node feature vector and pass through a feed-forward neural network (FFN) for nonlinear transformation, obtaining the final hidden representation of the target node at time t :

$$\tilde{h}_0^{(l)}(t) = \text{FFN}(h(t) \parallel x_0) = \text{Relu}([h(t) \parallel x_0] W_0^{(l)} + b_0^{(l)}) W_1^{(l)} + b_1^{(l)} \quad (11)$$

where FFN parameters capture nonlinear relationships between aggregated neighbor representation and target node features.

7. Fully Connected Layer

After the GAT layer, input learned node features into a fully connected layer to obtain the final output result:

$$\text{output} = \sigma(W_0 \cdot \tilde{h}_0^{(<ct>I<ot>)} + b) \quad (12)$$

where W_0 is the fully connected layer weight matrix, and b is the bias term.

So, The network architecture parameters are as shown in the Figure 3.

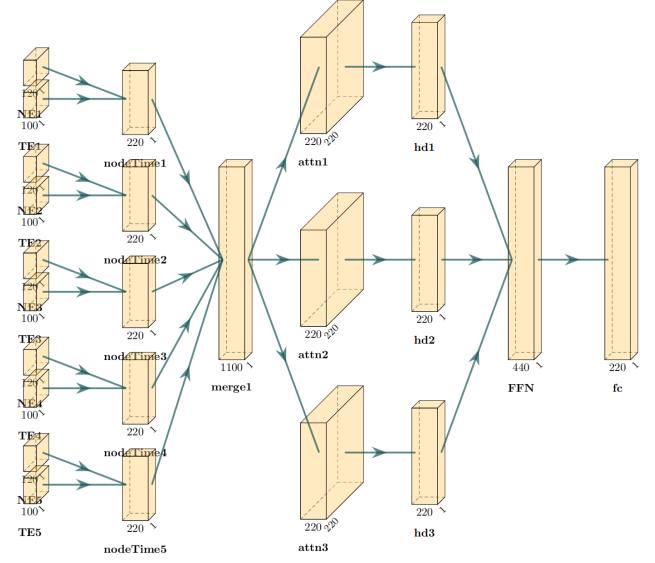


Figure 3: Network architecture parameters

Table 1: Dataset Division

Dataset	node	edge
Train	15819	23254
Valid	1751	2908
Test	2205	3021

3 Experiment and Result Analysis

3.1 Dataset

The dataset used in this experiment comes from a city bank's transaction fraud project. It contains 19,775 transaction events involving 7,497 accounts, with 1,102 suspected fraudulent users and 6,395 normal accounts. The dataset includes fields such as transaction event ID, transaction time, initiating user, recipient user, transaction amount, event type, summary, etc. The dataset is divided into training, validation, and test sets in an 8:1:1 ratio. The size of each subset is shown in Table 1.

3.2 Experimental Environment

The network model was built using the PyTorch framework and trained on an NVIDIA GeForce RTX 4090 GPU.

3.3 Evaluation Metrics

In fraud detection, common evaluation metrics include AUC, F1 score, and KS value. The KS value (Kolmogorov-Smirnov statistic) measures the distinction ability between positive and negative samples by calculating the maximum vertical distance between the cumulative distribution functions (CDF) of positive and negative samples. Mathematically, it is expressed as:

$$KS = \max(|F_{TPR}(t) - F_{FPR}(t)|) \quad (13)$$

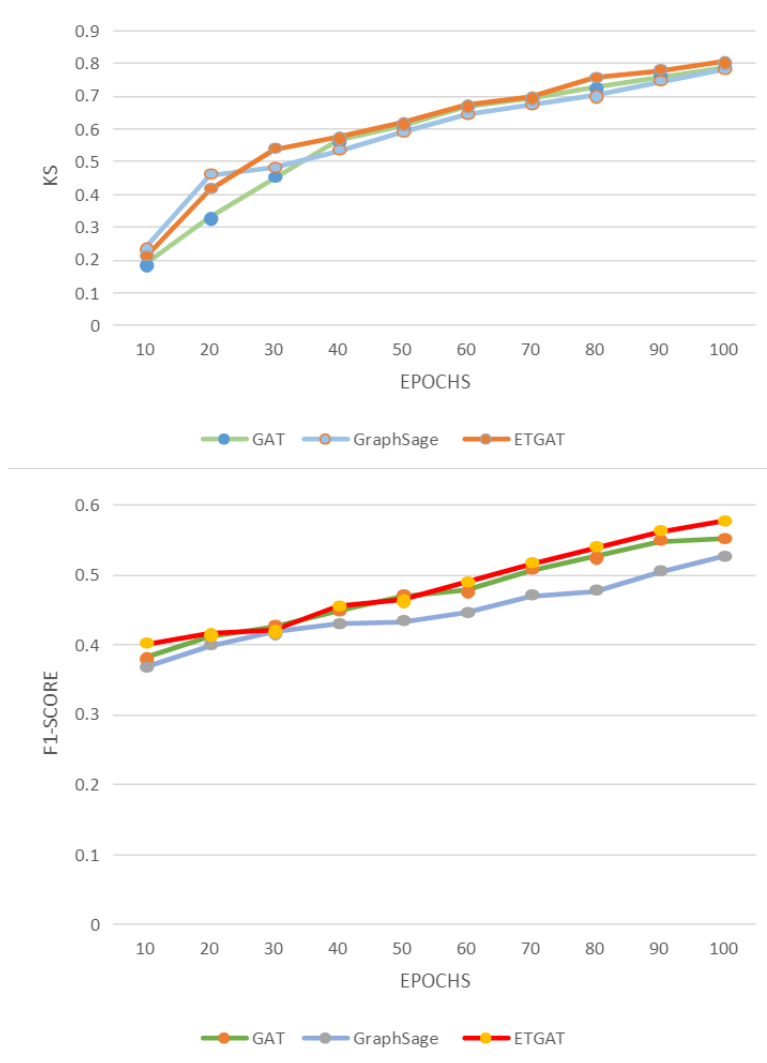


Figure 4: (a) KS Comparison on the Training Set (b) F1-SCORE Comparison on the Training Set

Table 2: Performance Comparison of Various Models

Model	AUC	KS	F1-SCORE
Graph SAGE	0.9518	0.7814	0.5257
GAT	0.9523	0.7861	0.5514
ETGNN	0.961	0.806	0.5726

where F_{TPR} is the true positive cumulative rate, and F_{TPR} is the false positive cumulative rate.

Figure 3 shows the comparison of KS values on the training set for algorithms GAT, GraphSage, and ETGNN, indicating that ETGNN has better distinction ability. Figure 4 shows the comparison of F1-SCORE on the training set for GAT, GraphSage, and ETGNN, showing that ETGNN also performs well in F1-SCORE.

On the test set, the performance comparison of various models is shown in Table 2, demonstrating that ETGNN outperforms the other two typical deep graph algorithms in all metrics.

4 Conclusion

In the field of transaction fraud detection, achieving high accuracy is often challenging. In real-world applications, incorrect predictions

can lead to serious consequences, such as transaction failures or account bans. Therefore, high precision is required, and recall should be maximized under this premise. Additionally, the model's predictions need to be interpretable to facilitate business troubleshooting and model iteration. The proposed event-temporal graph neural network primarily addresses the shortcomings of traditional user-centric graph construction methods. By introducing event-based graph construction and time encoding, it effectively learns unique operations in transaction fraud or money laundering, significantly improving recognition accuracy. However, as the number of transaction events accumulates over time, it becomes impractical to construct graphs for all events. For practical business applications, higher requirements must be met for graph construction and model prediction, balancing model complexity and time complexity to ensure reliable improvements in actual business operations.

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